SECTION TABLE OF CONTENTS

DIVISION 03 - CONCRETE

SECTION 03 20 00

CONCRETE REINFORCING

-- End of Section Table of Contents --

SECTION 03 20 00

CONCRETE REINFORCING

PART 1   GENERAL

1.1   REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN CONCRETE INSTITUTE (ACI)

ACI 318/318M	(2019; R 2022) Building Code Requirements for Structural Concrete (ACI 318-19) and Commentary (ACI 318R-19)
ACI SP-66	(2004) ACI Detailing Manual

ASTM INTERNATIONAL (ASTM)

ASTM A1034/A1034M-10a	(2015) Standard Test Methods for Testing Mechanical Splices for Steel Reinforcing Bars
ASTM A1064/A1064M	(2018a) Standard Specification for Carbon-Steel Wire and Welded Wire Reinforcement, Plain and Deformed, for Concrete
ASTM A184/A184M	(2006; E2011) Standard Specification for Fabricated Deformed Steel Bar Mats for Concrete Reinforcement
ASTM A276	(2010) Standard Specification for Stainless Steel Bars and Shapes
ASTM A370	(2020) Standard Test Methods and Definitions for Mechanical Testing of Steel Products
ASTM A615/A615M	(2020) Standard Specification for Deformed and Plain Carbon-Steel Bars for Concrete Reinforcement
ASTM A706/A706M	(2016) Standard Specification for Low-Alloy Steel Deformed and Plain Bars for Concrete Reinforcement
ASTM A767/A767M	(2016) Standard Specification for Zinc-Coated (Galvanized) Steel Bars for Concrete Reinforcement
ASTM A780/A780M	(2020) Standard Practice for Repair of

	Damaged and Uncoated Areas of Hot-Dip Galvanized Coatings
ASTM A955/A955M	(2012; E 2012) Standard Specification for Deformed and Plain Stainless-Steel Bars for Concrete Reinforcement

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Designation RO stands for Resident Office and designation DO stands for District Office. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Reinforcement; G, DO

Detail drawings showing reinforcing steel placement, schedules, sizes, grades, and splicing and bending details. Drawings shall show support details including types, sizes and spacing.

SD-03 Product Data

Materials; G, DO

Mechanical Butt-Splices; G, DO

SD-06 Test Reports

Tests, Inspections, and Verifications; G, DO

Test Butt-Splices; G, DO

SD-07 Certificates

Reinforcing Steel; G, DO

Qualification of Steel Bar Butt-Splicers; G, DO

1.3 QUALITY ASSURANCE

1.3.1 Qualification of Steel Bar Butt-Splicers

Personnel installing steel bar butt-splices shall have satisfactorily completed a course of instruction in the proposed method of butt-splicing or have satisfactorily performed such work within the preceding year. Submit certificates on the Qualifications of Steel Bar Butt-Splicers prior to commencing butt-splicing.

1.3.2 Qualification of Butt-Splicing Procedure

As a condition of approval of the butt-splicing procedure, make three test butt-splices of steel bars of each size to be spliced using the proposed butt-splicing method, in the presence of the Contracting Officer. Tension test to ultimate tensile strength these test butt-splices and unspliced bars of the same size, and plot stress-strain curves for each test. Test results must show that the butt-splices meet the specified strength and deformation requirements in order for the splicing procedure to be approved.

1.4 DELIVERY, STORAGE, AND HANDLING

Store reinforcement and accessories off the ground on platforms, skids, or other supports. Protect mechanical splice threads from rusting.

PART 2 PRODUCTS

2.1 STAINLESS STEEL DOWELS

Conform to ASTM A955/A955M or ASTM A276; 316 Series UNS S31600, UNS S31603, UNS S31653 or UNS S 31803, as specified or shown, for stainless steel dowels. All stainless steel dowels shall be Grade 60, minimum yield strength of 60,000 psi.

2.2 GALVANIZED DOWELS

Galvanized dowel bars are deformed bars conforming to ASTM A706/A706M and ASTM A767/A767M. All galvanized dowels shall be Grade 60, minimum yield strength of 60,000 psi. Provide a Class 1 coating and fabricate dowel bars prior to galvanizing.

Repair damaged surfaces with galvanizing repair method and paint conforming to ASTM A780/A780M or by application of zinc rich paint galvanizing, as approved by Contracting Officer. The length and width of the damaged surface area that shall be allowed is not to exceed 1 in. in only one of the two dimensions. Areas to be repaired shall be cleaned, dry, free of oil, grease, preexisting paint, and corrosion products. Hand clean the surface to the requirements of SSPC SP2. Apply zinc rich paints as soon as possible after surface preparation to avoid the development of any oxide. Follow the paint manufacturer's instructions and be aware of the recommended temperature, humidity conditions, curing time, and paint thickness per application. Spray or brush the zinc paint onto the surface with horizontal passes per the manufacturer's recommendations. Once the repair coating has cured, the product can be inspected and the thickness verified. The final coating thickness for areas repaired with the zinc rich paint should be 50% more than the galvanized coating thickness requirements as specified in ASTM A767/A767M, but not more than 4.0 mils. Multiple applications of zinc rich paint may be necessary to achieve coating thickness repair requirements. Follow the manufacturer's curing instructions between applications.

2.3 FABRICATED BAR MATS

Fabricated bar mats conforming to ASTM A184/A184M.

2.4 REINFORCING STEEL

Reinforcing steel of deformed bars conforming to ASTM A615/A615M or ASTM A706/A706M as indicated. Welding of any reinforcement is not permitted. When the grade of steel is not specified in the plans, Grade 60 shall be provided. Cold drawn wire used for spiral reinforcement must conform to ASTM A1064/A1064M.

Submit certified copies of mill reports attesting that the reinforcing steel furnished contains no less than 25 percent recycled scrap steel and meets the requirements specified herein, prior to the installation of reinforcing steel. Certified mill reports shall not be older than 6 months.

2.4.1 Mechanical Butt-Splices

Mechanical butt splices must be exothermic, threaded coupling, swaged sleeve or other positive connecting type, and shall develop in tension or compression at least 1.25 times the yield strength of the bar as well capable of developing the specified tensile strength of the spliced bar according to ACI 318/318M, Type 2. In addition to this strength requirement, the additional deformation of number 14 and smaller bars due to slippage or other movement within the splice sleeve shall not exceed 0.015 inches (unit strain 0.0015 inches/inch) beyond the elongation of an unspliced bar based upon a 10 inch gage length spanning the extremities of the sleeve at a stress of 30,000 psi. The additional deformation of number 18 bars must not exceed 0.03 inches (unit strain 0.003 inches/inch) beyond the elongation of an unspliced bar based upon a 10 inch gage length spanning the extremities of the sleeve at a stress of 30,000 psi. Determine the amount of the additional deformation from the stress-strain curves of the unspliced and spliced bars tested as required in paragraph QUALIFICATION OF BUTT-SPLICING PROCEDURE for qualification of the butt-splicing procedure.

2.5 WELDED WIRE REINFORCING

Welded wire reinforcing conforming to ASTM A1064/A1064M. For wire with a specified yield strength (fy) exceeding 60,000 psi, fy must be the stress corresponding to a strain of 0.35 percent.

2.6 WIRE TIES

Use wire ties that are 16 gauge or heavier black annealed steel wire.

2.7 SUPPORTS

Provide bar supports complying with the requirements of ACI SP-66. Provide plastic-coated wire, stainless steel or precast concrete supports for bars in concrete with formed surfaces exposed to view or to be painted. Use wedge-shaped precast concrete supports, not larger than 3-1/2 by 3-1/2 inches, of thickness equal to that indicated for concrete cover and with an embedded hooked tie-wire for anchorage. Bar supports used in precast concrete with formed surfaces exposed to view must be the same quality, texture and color as the finish surfaces.

2.8 TESTS, INSPECTIONS, AND VERIFICATIONS

Perform material tests, specified and required by applicable standards, by an approved laboratory and certified to demonstrate that the materials are in conformance with the specifications. Perform and certify tests, inspections, and verifications. Submit certified tests reports not older than 6 months of reinforcement steel showing that the steel complies with the applicable specifications for each steel shipment and identified with specific lots prior to placement. Submit three copies of the heat analyses for each lot of steel furnished certifying that the steel conforms to the heat analyses.

2.8.1 Reinforcement Steel Tests

Perform mechanical testing of steel (one test for every ~~5~~100 tons per lot of steel) in accordance with ASTM A370 except as otherwise specified or required by the material specifications. Perform tension tests on full cross-section specimens using a gage length that spans the extremities of specimens with welds or sleeves included. From chemical analyses of steel heats, report the percentages of carbon, phosphorous, manganese, sulphur and silicon present in the steel.

PART 3 EXECUTION

3.1 REINFORCEMENT

Fabricate and place reinforcement steel and accessories as specified, as indicated, and as shown on approved shop drawings. Fabrication and placement details of steel and accessories not specified or shown must be in accordance with ACI SP-66 and ACI 318/318M. Cold bend reinforcement unless otherwise authorized. Bending may be accomplished in the field or at the mill. Do not bend bars after embedment in concrete. Place safety caps on all exposed ends of vertical concrete reinforcement bars that pose a danger to life safety. Face wire tie ends away from the forms. Submit detail drawings showing reinforcing steel placement, schedules, sizes, grades, and splicing and bending details. Show support details including types, sizes and spacing.

3.1.1 Placement

Reinforcement must be free from loose rust and scale, dirt, oil, or other deleterious coating that could reduce bond with the concrete. Place reinforcement at locations indicated plus or minus one bar diameter. Do not continue reinforcement through isolation joints and place as indicated through construction, expansion, or contraction joints. Cover with concrete coverage as indicated in plans. If bars are moved more than one bar diameter to avoid interference with other reinforcement, conduits or embedded items, the resulting arrangement of bars, including additional bars required to meet structural requirements, requires approval before concrete is placed.

3.1.2 Placing Tolerances

3.1.2.1 Spacing

The spacing between adjacent bars and the distance between layers of bars may not vary from the indicated position by more than one bar diameter nor more than 1 inch.

3.1.2.2 Concrete Cover

The minimum concrete cover of main reinforcement steel bars shall be as shown. The allowable variation for minimum cover shall be as follows:

MINIMUM COVER (inch)	VARIATION (inch)
6	plus 1/2
4	plus 3/8
3	plus 3/8

MINIMUM COVER (inch)	VARIATION (inch)
2	plus 1/4
1-1/2	plus 1/4
1	plus 1/8
3/4	plus 1/8

3.1.3 Splicing

Conform splices of reinforcement to ACI 318/318M and make only as required or indicated. Bars may be spliced at alternate or additional locations at no additional cost to the Government subject to approval. Splicing must be by lapping or by mechanical splices; except that lap splices must not be used for bars larger than No. 11 unless otherwise indicated.

3.1.3.1 Lap Splices

Place lapped bars in contact and securely tied or spaced transversely apart to permit the embedment of the entire surface of each bar in concrete. Do not space lapped bars farther apart than 1/5 the required length of lap or 6 inches. Lap splices shall be staggered so that no more than half of the bars are spliced in any one section unless otherwise indicated.

3.1.3.2 Mechanical Butt-Splices

Fabricate mechanical butt-splices in accordance with the mechanical splicing device manufacturer's recommendations as well as ASTM A615/A615M or ASTM A706/A706M, Grade 60. Bars to be spliced by a mechanical butt-splicing process may be sawed or sheared, provided the ends of sheared bars are reshaped after shearing. Clean surfaces to be enclosed within a splice sleeve or coupling by wire brushing or other approved method prior to splicing. Make splices using manufacturer's standard jigs, clamps, and other required accessories. Longitudinally stagger tension splices of number 14 or smaller bar a minimum of 5 feet or as otherwise indicated so that no more than half of the bars are spliced at any one section. Longitudinally stagger tension splices of number 18 bars a minimum of 5 feet so that no more than 1/3 of the bars are spliced at any one section. In addition, splices in adjacent bars shall be staggered at least 30 inches.

3.2 DOWEL INSTALLATION

Install dowels at locations indicated on drawings and at right angles to joint being doweled. Accurately position and align dowels parallel to the finished concrete surface before concrete placement. Rigidly support dowels during concrete placement. Coat one end of dowels with a bond breaker, asphalt, grease or plastic sleeve to prevent a bond with the concrete as noted on drawings.

3.3 FIELD TESTS AND INSPECTIONS

3.3.1 Identification of Splices

Establish and maintain an approved method of identification of all field butt-splices which will indicate the splicer and the number assigned each splice made by the splicer.

3.3.2 EXAMINING, TESTING, AND CORRECTING

Perform the following during the butt-splicing operations as specified:

3.3.2.1 Tension Tests

Prepare butt-splicing test specimens and test in accordance with ASTM A1034/A1034M-10a to the specified ultimate tensile strength of the spliced bars or to destruction. The number of tests to be performed are as follows:

- Six(6) tension tests for the bridge middle pier shaft splice specimens selected by the Contracting Officer for the approval of the splicing procedure.

- Three(3) tests per each sampling lot for channel wall shafts. Splice specimens shall be selected by the Contracting Officer per each sampling lot for approval of the splicing procedure and 10 additional specimens selected at the discretion of the Contracting Officer during installation.

Each sampling lot is defined as 500 splicers (or fraction thereof) from the same coupler producer and of the same nominal size, length, and grade.

3.3.2.2 Test Specimens

Butt-splicing test specimens shall be assembled with the same type, kind, and rating of equipment or tools intended to be used for the production butt-splices. Provide minimum bar lengths on each side of butt-splice as follows:

- 19-inch for #9 bars and smaller
- 24-inch for #10 bars and larger

3.3.2.3 Correction of Deficiencies

No splice shall be embedded in concrete until satisfactory results of visual examination and the required tests or examinations have been obtained. All splices having visible defects or represented by test specimens which do not satisfy the tests or examinations shall be removed. If any of the tension test specimens fail to meet the strength requirements or deformation limitations two production splices from the same lot
represented by the test specimens which failed shall be cut out and tension tested. If both of the retests pass the strength requirements and deformation limitations all of the splices in the lot will be accepted. If one or both of the retests fail to meet the strength requirements or deformation limitations all of the splices in the lot will be rejected. All costs of removal, testing and resplicing of the additional production splices shall be borne by the Contractor. The bars of rejected splices shall be cut off outside the splice zone of coupling or sleeve. The cut

ends shall be finished as specified and the joints shall be respliced and reinspected at no additional cost.

3.3.2.4 Supplemental Examination

The Contracting Officer may require additional or supplemental radiographic examination and/or tension test of any completed splice.

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